AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M238
Data Visualization

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 09/April/2025

AOU EGYPT

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed:** (1.5) Hours

Instructions:

- Total Marks: **30**
- This exam consists of **Three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

CamScanner

PART 1

TRUE or FALSE (5 marks)

MARK the following statements are TRUE or FALSE

1. Visualization is suitable when there is a need to augment human capabilities rather than replace them with computational methods.
2. Visualization designers must take into account three very different kinds of resource limitations: those of computers, of humans, and of displays.
3. Qualitative color, sequential color, and diverging color are examples of color scales.
4. Humans excel at detecting patterns in raw numerical data without the need for visual aids.
5. In a mosaic plot both the heights and the widths of individual shaded areas are similar.

PART 2

ALL QUESTIONS ARE REQUIRED (19 marks)

1. What are two types of visualization approaches? And explain each type. (4 Marks)
2. Differentiate between Ugly, bad, and wrong figures. (6 Marks)
3. Describe the importance of having a human in the loop for data visualization. (2 Marks)
4. Define Heatmaps ? (1 mark)
5. List the three fundamental use cases for color in data visualizations? (3 marks)
6. List six commonly used aesthetics in data Visualization. (3 Marks)

PART 3

ALL QUESTIONS ARE REQUIRED (6 marks)

1. Write a Python code for figure 1, to load the housing dataset using Pandas and create a line plot showing the 'Year' versus 'Average_Price'. (6 marks)

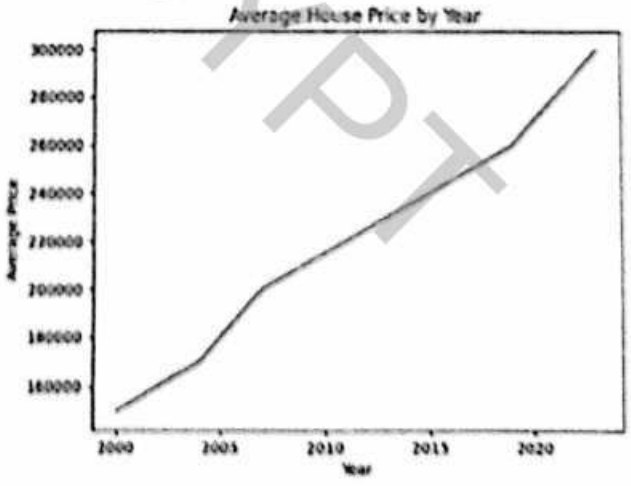


Figure 1

End of questions

KOLEGY PT

CamScanner